

Ankit Mukherjee, MSc

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🌐 AnkitMukherji

Summary

Computational biologist with research experience spanning **population genomics, pharmacogenomics, transcriptomics, and viral evolution**. My work focuses on leveraging large-scale genomic and transcriptomic datasets to understand **human genetic diversity, disease mechanisms, and precision medicine**. At CSIR–Institute of Genomics and Integrative Biology, I have contributed to the GenomeIndia Project through the characterization of clinically actionable pharmacogenomic variation across diverse Indian populations, development of genomic resources, and transcriptomic analysis of Friedreich Ataxia disease models. My long-term research interests lie in integrating computational genomics with molecular biology to investigate disease biology, gene regulation, and complex human disorders.

Education

Savitribai Phule Pune University <i>MSc in Bioinformatics (DBT funded)</i> Grade: 9.93/10	Pune 2022–2024
University of Calcutta <i>BSc in Microbiology</i> Grade: 9.03/10	Kolkata 2019–2022
St. Augustine's Day School <i>Higher Secondary (Class XII, Science)</i> Percentage: 95.5%	Barrackpore 2018–2019
St. Augustine's Day School <i>Secondary (Class X)</i> Percentage: 92.4%	Barrackpore 2016–2017

Research Experience

CSIR–Institute of Genomics and Integrative Biology <i>Transcriptomics of Friedreich Ataxia</i>	Delhi 2025–Present
- Investigating transcriptomic alterations underlying Friedreich Ataxia (FRDA) using both patient blood RNA-seq datasets and induced pluripotent stem cell (iPSC)-derived cardiomyocyte models. - Performed differential expression and pathway enrichment analyses to identify blood-based transcriptomic biomarkers associated with disease severity, with the long-term objective of developing non-invasive molecular markers for patient stratification and disease monitoring. - Designed reproducible RNA-seq analysis pipelines involving quality assessment, normalization, batch-effect correction, differential expression analysis and functional enrichment using Bioconductor, DESeq2, limma and custom R workflows. - Currently investigating transcriptional remodeling during differentiation of Day 15 and Day 52 iPSC-derived cardiomyocytes from Friedreich Ataxia patients to understand molecular mechanisms contributing to progressive cardiomyopathy, the leading cause of mortality in FRDA.	
CSIR–Institute of Genomics and Integrative Biology (GenomeIndia Project) <i>Pharmacogenomics and Medical Relevance of Indian Populations</i>	Delhi 2024–2026
- Investigating the landscape of clinically actionable pharmacogenomic variation across the Indian population by integrating whole-genome sequencing data from 10,000 individuals representing 83 genetically diverse populations. - Curated and harmonized pharmacogenomic evidence from PharmGKB, CPIC, PharmVar, ClinPGx and additional clinical resources to identify actionable SNPs, indels and star alleles relevant for precision medicine. - Characterized population-specific pharmacogenomic variation, quantified inter-population allele frequency differences, and compared Indian populations with global reference datasets to identify ancestry-specific therapeutic implications. - Performed comprehensive analyses of pharmacogenes including clinically important star alleles and metabolizer phenotypes to evaluate differences in predicted drug response among Indian populations. - Developed computational pipelines in R, Python and Bash for large-scale variant annotation, quality control, population-level analyses, visualization and statistical interpretation.	

National Centre for Cell Science (NCCS)

Master's Thesis

Pune

2023–2024

- Analyzed **viral genomes** to identify amino acid substitutions under positive selection, revealing sites of rapid evolution.
- Constructed custom **substitution matrices** based on the PAM algorithm and compared them to standard **PAM matrices** to identify amino acid mutations under selection pressure.
- Mapped key mutations onto protein structures, demonstrating a **non-random** distribution of substitutions localized to specific functional regions that facilitate the host-pathogen evolutionary arms race.

National Centre for Cell Science (NCCS)

Summer Research

Pune

2023

- Characterized the impact of ethnicity-related SARS-CoV-2 SNPs on glycosylating proteins to unravel their role in modulating viral susceptibility across diverse populations.
- Performed **docking** analyses to assess how specific SNPs influence infection susceptibility, demonstrating a direct link between genetic variation and viral binding.

Research Resources

GI-DB: Contributing author of **GI-DB (GenomeIndia Database)**, a publicly accessible genomic resource cataloguing genetic variation from **10,000 whole genomes representing 83 Indian populations**. Contributed to large-scale variant processing, quality assessment, annotation and population-level analyses supporting the development of a national reference database for Indian genomic variation.

Database: ibdc.dbt.gov.in/gidb

Publications

- Bhattacharyya, C., Subramanian, K., Uppili, B., ..., **Mukherjee, A.**, et al. (2025). *Mapping Genetic Diversity with the GenomeIndia Project*. *Nature Genetics*, 57, 767–773. DOI:10.1038/s41588-025-02153-x
- Subramanian, K., Bhattacharyya, C., **Mukherjee, A.**, et al. (2026). *An Atlas of Indian Genetic Diversity*. *medRxiv*. Preprint. DOI:10.64898/2026.03.20.26348801

Skills

Programming & Scripting: Python (NumPy, Pandas), R (Tidyverse, Bioconductor, ggplot2), Shell (awk, sed), Perl

NGS & Genomics: Variant Calling (GATK, BCFtools), RNA-seq (Kallisto, DESeq2, Seurat, Scanpy), Genome/Exome Assembly, Functional Annotation

Computational Tools: High-Performance Computing (HPC, SLURM), Version Control (Git), Containerization (Docker, Singularity)

Research & Soft Skills: Scientific Communication, Data Visualization, Problem Solving, Collaborative Research

Achievements

First Prize – Best e-Poster Presentation, Indian Ataxia Conclave, for presenting research on transcriptomic biomarkers in Friedreich Ataxia.

Qualified DBT-BET 2024 (Category-II, 214 marks)

AIR 51 – Graduate Aptitude Test in Biotechnology (GAT-B 2022)

AIR 192 – IIT-JAM Biotechnology 2022

Qualified TIFR GS 2022 (Biology stream)

Best Student Award – BSc Microbiology, Calcutta University

Languages

English: Professional proficiency

Hindi: Professional proficiency

Bengali: Native